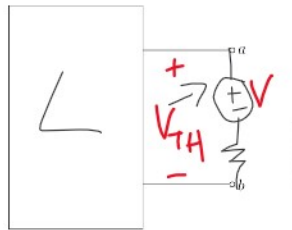


L15 - Thevenin's Theorem

Thursday, March 14, 2024 11:09 AM

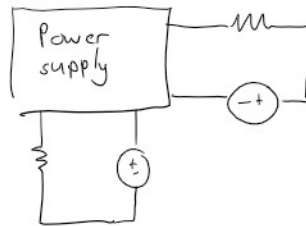
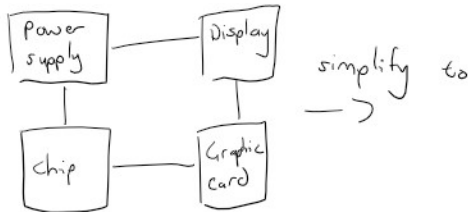
Core Principles

- V_{Th} is voltage across terminals
- R_{Th} is equivalent resistance across terminals
 - Turn off all independent sources
- If there is a dependant voltage source, must add a test source (1V or 1A)
 - $R_{Th} = V_{test} / I_{test}$



Why use thevenin and norton:

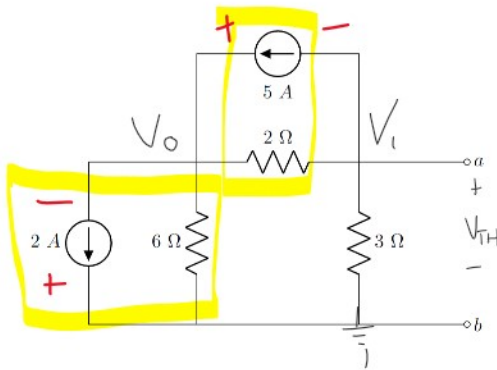
In large circuits with multiple systems. e.g motherboard



terminals a-b
polarity: + -

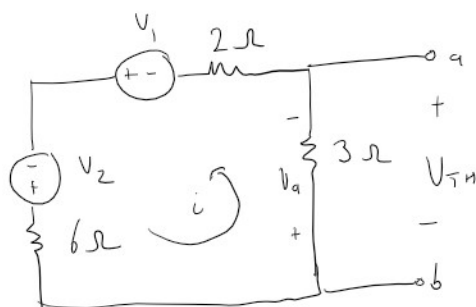
Steps:

- Find voltage across requested terminals to get V_{th}
- Turn off all independent sources
- Is there a dependent source?
 - If no, find equivalent resistance for R_{th}
 - If yes, draw a 1V or 1A source between terminals
 - Calculate the current through or voltage across that source
 - Use ohms law to find R_{th}



① Find V_{Th}

Using source transformation:



$$V_1 = 5(2) = 10V$$

$$V_2 = 2(6) = 12V$$

$$KVL @ i: 3i + 2i - 10 - 12 + 6i = 0$$

$$22 = 11i$$

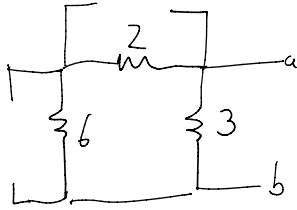
$$i = 2A$$

$$V_a = 3(i) = 6V$$

$$V_a = -V_{Th} \therefore V_{Th} = -6V$$

② find R_{TH} .

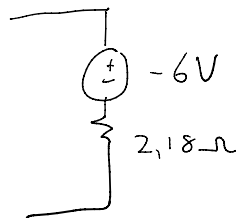
a) turn off independent sources



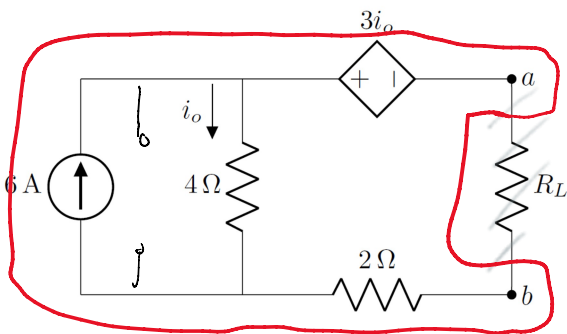
$$2 + 6 = 8 \Omega \text{ (series)}$$

$$\frac{8 \cdot 3}{8 + 3} = 2,18 \Omega \quad R_{TH} = 2,18 \Omega$$

$\therefore$ Thevenin eq is

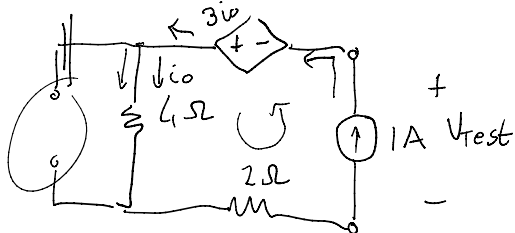


Example 2:



Find R_{th} in reference to terminals a-b across an unknown load R_L :

① turn off all ind sources



② Add 1A $\uparrow$ V $\oplus$

$$\textcircled{3} R_{TH} = \frac{V_{test}}{I_{test}} = V_{test}$$

$$\therefore i_o = 1A$$

KVL @ i_o

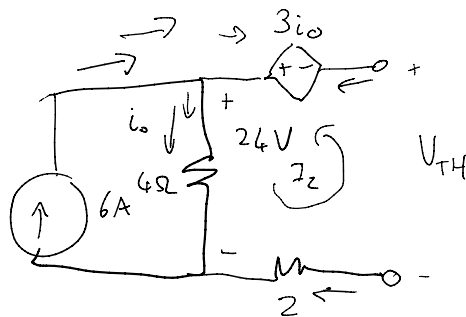
$$-V_{test} + (-3) + 4(i_o) + (2)(i_o) = 0$$

$$-V_{test} + (-3) + 4 \cdot 1 + 2 \cdot 1 = 0$$

$$-V_{test} = -3$$

$$V_{test} = 3V$$

$$\therefore R_{TH} = 3 \Omega$$



$$i_o = 6A$$

$$V_{TH} \neq 24V$$

KVL @ I_2

$$24 - V_{TH} - 3i_o = 0$$

$$V_{TH} = -3i_o + 24$$

$$= -18 + 24$$

$$V_{TH} = +6V$$